



The CIA Triad & Threat Landscapes

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What is Cyber Security?

Cyber Security is the practice of protecting systems, networks, programs, and data from digital attacks. It requires establishing multi-layered security controls across the computing infrastructure.

Core Concepts & Layers

- **Vulnerability:** A weakness or loophole in system code or configuration (e.g. unpatched software).
- **Threat:** A potential danger that could exploit a vulnerability (e.g. hacker group, malware).
- **Assets & Harm:** Assets are data, hardware, and services. Harm is the impact of a threat exploiting a vulnerability.
- **Defense in Depth:** Implementing security at multiple layers (Physical, Network, Host, Application, Data).

Motives & Attacker Profiles

- **Cyber Criminals:** Primarily motivated by financial gain (e.g. ransomware, credit card fraud).
- **Hacktivists:** Motivated by political or social causes (e.g. defacing websites, leaking documents).
- **Nation-State Agents:** Cyber warfare, espionage, and sabotaging foreign infrastructure.
- **Insider Threats:** Malicious or negligent employees leaking credentials from within.

The CIA Triad

The CIA Triad is the core model that guides security policies and controls:

1. Confidentiality

- Goal: Prevent unauthorized access to sensitive information.
- Violations: Data breaches, packet sniffing, unauthorized file reads.
- Controls: Encryption (AES), multi-factor authentication (MFA), access control lists.

2. Integrity

- Goal: Prevent unauthorized or undetected modification of data.
- Violations: Man-in-the-middle edits, database tampering, virus code injection.
- Controls: Hashing (SHA-256), digital signatures, file checksums.

3. Availability

- Goal: Guarantee reliable, timely access to systems for authorized users.
- Violations: DDoS attacks, server crashes, power grid failures.
- Controls: Redundant servers, load balancers, backups, DDoS protection.



Active vs. Passive Attacks

Understanding how threat actors interact with data determines our defensive controls:

Characteristic	Passive Attacks	Active Attacks
Core Action	Monitoring, sniffing, and reading traffic	Modifying, blocking, or injecting fake data
System Impact	No alteration of database or system state	Alters data, system states, or network operations
Examples	Packet sniffing, traffic analysis, eavesdropping	DDoS, man-in-the-middle, replay, spoofing
Detection Style	Hard to detect (requires packet audits)	Easy to detect (crashes or alerts trigger)
Primary Defense	Encryption (hides readable content)	Firewalls, intrusion detection, authentication

Applications of Security Policies

How organizations deploy the CIA triad and attack countermeasures in practice:

Multi-Factor Authentication (MFA)

Protects Confidentiality. Requires users to provide two or more verification factors (password + SMS token + biometric fingerprint) to login.

Intrusion Detection Systems (IDS)

Protects Availability and Integrity. Actively scans incoming network packets for signatures of active attacks, blocking malicious IPs automatically.

End-to-End Encryption (E2EE)

Protects Confidentiality and Integrity. Encrypts data at the source and decrypts only at destination, preventing passive packet sniffing mid-transit.

System Backup & DR plans

Protects Availability. Regularly clones database systems to offline locations, ensuring quick recovery in the event of a ransomware attack.

THANK YOU

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